

When Calibration Is Not Enough: A Documented Deployment Gap for Camera-Driven Underwater Obstacle Avoidance

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Abstract. We investigate why calibrating measured sensing and actuation did not yield closed-loop transfer for camera-driven underwater obstacle avoidance. A HoloOcean baseline was augmented with observation, timing and vehicle-response models fitted to a BlueROV2. Eighty closed-loop simulations covering two planners and two obstacle geometries were frozen, hashed and committed before the physical campaign (PRE). The deployed stack required documented changes; nine runs were then acquired and eight were admissible, all with the committed planner (REAL). No overhead tracking was recorded during these runs, so spatial metrics such as true clearance and trajectory cannot be validated. The comparable command stream nevertheless shows that all eight physical runs remained at the planner’s strafe setpoint, whereas all twenty corresponding frozen runs reached its go-around setpoint. The principal mismatches were structural: the detector accepted the obstacle only from 0.29 m to 2.11 m; the vehicle started 1.86 m away and engaged at 1.80 m; and a 2.5 m clearing offset could not fit the 2.7 m to 3.0 m-wide basin. Eighty further simulations performed only after deployment (POST) isolate these effects. Correcting a relay field-of-view defect improved simulated clearance, but the tested pool-feasible manoeuvre rescaling was unsafe in four of ten runs. Thus, calibrating measurable sensing and actuation mismatches did not ensure transfer when deployment was governed by structural mismatches absent from the model.

Keywords: Sim-to-real transfer · Underwater obstacle avoidance.

1 Introduction

Underwater autonomy is commonly developed in simulation and judged in water. The resulting deployment gap is often reported as a change in success rate or error, without identifying its mechanism. We document that mechanism for one camera-driven obstacle-avoidance stack, one vehicle and one pool.

We tested whether measuring simulator mismatches improves closed-loop transfer. Three models were fitted to the physical system: a static observation model, a timing and burst-dropout model, and a vehicle model covering saturation, deadband, asymmetry and lag. The vehicle model acts downstream of the planner, so calibration never edits the controller. Together with the baseline, these form levels S0–S3.

Before the vehicle entered the water, we executed, verified, hashed and committed 80 closed-loop runs: four levels, two planners, two obstacle geometries and five repetitions. These are predictive evidence (PRE), not reconstructions informed by the deployment.

The frozen stack did not act on the vehicle despite frequent detections. Seven changes were made during the only available pool day. Nine runs were then recorded with the committed planner and eight are admissible (REAL). The dynamic-window baseline could not be used because the vehicle lacks the pose estimate it requires. The physical dataset is closed.

Because the deployed and frozen stacks differ, the physical runs cannot establish causal simulator predictivity or compare planners. They can document what changed and motivate a post-deployment diagnosis.

Two structural mismatches dominate. Detector size gates create an observable range window of 0.29 m to 2.11 m. The vehicle started 1.86 m from the anchor and engaged at 1.80 m, eliminating the simulated approach phase. The planner also requests a 2.5 m lateral offset and a 4 m run-past manoeuvre in a basin only 2.7 m to 3.0 m wide. Neither mismatch was represented by S1–S3.

No external tracking was recorded during the runs. Trajectory, path length, lateral excursion and true clearance therefore cannot be recovered without assuming the plant behaviour being evaluated. We use only the recorded command stream and the observation stream recovered from the detector’s annotated video. Spatial comparisons are excluded.

The final audit also found a field-of-view inconsistency in the observation relay and a discarded random seed. We retain the frozen results and report these defects explicitly.

Our contributions are:

1. a measurement-grounded S0–S3 calibration ladder, together with an 80-run pre-deployment campaign with committed provenance;
2. a file-level account of the seven deployment changes and the structural mismatches that governed the real campaign;
3. an explicit command-domain comparison, with the boundary imposed by missing spatial ground truth; and
4. an 80-run post-deployment ablation that separates diagnostic evidence from the frozen predictions.

Evidence frozen before deployment is marked PRE, physical evidence REAL, and analysis performed after observing deployment POST. Figure 1 makes this chronology explicit.

2 Related Work

2.1 Simulation fidelity and sim-to-real evaluation

HoloOcean provides the physics and rendering used in this study [7,8]; OceanSim reflects the broader effort to support underwater perception in simulation [9].

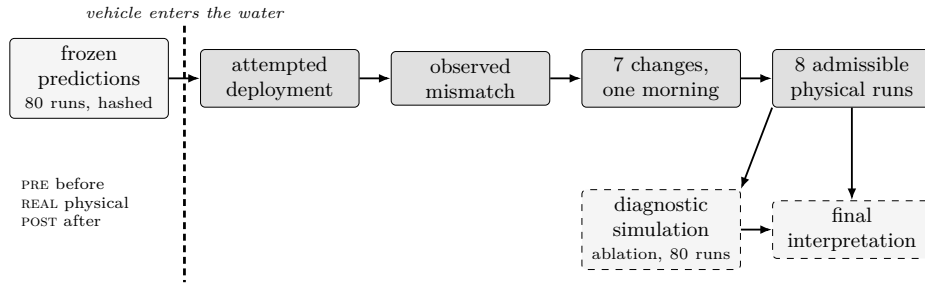


Fig. 1. The chronology of evidence, and the reason the paper marks it. Everything left of the rule was fixed before any physical observation existed and can be read as a prediction. Everything right of it, the diagnostic simulations included, was produced by people who already knew how the physical runs had gone. The eight physical runs feed the final interpretation directly and also motivate the diagnostic study, which is why that study can explain the deployment gap but cannot confirm the frozen predictions.

Simulation quality, however, is not equivalent to real-world predictivity. Kadian et al. formalise sim-to-real predictivity, while Truong et al. show that higher fidelity need not improve transfer [5,10]. Underwater work has examined direct high-fidelity transfer and HoloOcean-based hardware-in-the-loop evaluation [14,6]. We instead freeze predictions before deployment and preserve the later diagnosis as a separate evidence class.

2.2 Vision-based underwater avoidance

Attenuation, turbidity and uncertain monocular depth complicate underwater vision. Relevant approaches include learned depth estimation [13], navigation combining monocular vision with sonar and domain randomisation [11], and direct vision-based avoidance [2,1,12]. Our focus is not detector accuracy but the interval over which detector acceptance gates produce any planner-usable observation.

2.3 Scope of the contribution

The dynamic-window approach is a standard local baseline [4]; an underwater adaptation is given by Eriksen et al. [3]. We compare it with a committed-maneuvre controller only in simulation because the physical vehicle supplied no pose estimate for the dynamic-window planner. We claim no new planner, detector or simulator. The contribution is an experimentally traced deployment gap, including negative and post-hoc evidence, rather than an algorithmic advance.

3 System and the Perception-to-Actuation Pipeline

3.1 Vehicle and basin

The platform is an eight-thruster BlueROV2 Heavy running ArduSub over a tether, with a forward 1080px camera. It has neither Doppler velocity log nor acoustic positioning and therefore no absolute or velocity-referenced pose. The basin is a rectangular pool of roughly 6 m×2.7 m to 3.0 m, with the obstacle – a 0.5 m-class anchor – suspended from a transverse rod.

3.2 The pipeline and the boundary that both domains share

The stack is a chain of ROS 2 nodes, and the same source code runs in both domains between two fixed points:

```
/perception/obstacles_raw → qualification → estimator
                               → planner → /planner/cmd_vel_safe.
```

The source code within this boundary is identical in both domains. Raw observation producers and command consumers differ. This shared boundary defines the command-domain comparison; Section 6 documents the deployment changes.

Qualification requires mutually coherent updates before declaring a valid track, suppressing single-frame false positives at the cost of delay.

Estimation uses a fixed-gain Kalman filter on image position and extent and predicts through observation gaps. The same variant is fixed for every run.

Range is monocular and deliberately simple. An object of assumed physical height H whose bounding box fills a fraction h of the image subtends $\theta = h \cdot vfov$, giving

$$R = \frac{H/2}{\tan(\theta/2)}. \quad (1)$$

Both planners use this implementation and the same assumed height. Their range inversion used the 90° node default in both domains; the frozen observation relay alone used 60° when applying its detection-probability model (Section 10).

3.3 The two planners

The *committed* planner is a finite-state controller. Below an engagement range it commits to one side, holds that side for a minimum duration, strafes to a clearing offset, runs past the obstacle and returns to the original line. Its side choice depends only on where the obstacle appears in the image, which is the property Section 7 tests. Its manoeuvre geometry – the clearing offset and the run-past margin – is the pair of parameters that turns out not to fit the basin.

The *holonomic dynamic-window* baseline [4] scores velocity candidates for clearance, progress, speed, route adherence and smoothness. It requires a pose estimate to project candidates and evaluate route adherence.

3.4 Actuation

Simulated commands pass to a vehicle-dynamics model in HoloOcean. Physical commands map to ArduSub joystick channels through

$$c = \text{sign}(v) \left(d + \frac{|v|}{k} \right), \quad (2)$$

where per-direction gain k and deadband d are fitted from open-water trials. S3 applies the fitted response downstream of the planner.

4 Measurement-Grounded Calibration: the S0–S3 Ladder

4.1 Design principle: one owner per phenomenon

The ladder adds one measured phenomenon at a time, and each level owns exactly one. The rule matters because the obvious alternative – injecting everything measured at every level – double-counts: a detection-probability model and a dropout-burst model both reduce the number of observations that reach the planner, and applied together they reduce it twice. We therefore fix that S1 owns the *marginal* probability of detection and S2 owns only the *correlation structure* of the misses, parameterised so as to reproduce S1’s marginal exactly. The safeguard is structural rather than a matter of care: the two-state visible/blind gate is constructed so that its stationary visible fraction equals the marginal by identity, so S2 changes when the misses fall and never how many there are.

4.2 S1: a static observation model

S1 was fitted to paired samples in which the real detector’s output and an independently measured range to the anchor were recorded together ($n = 24$ usable pairs). Three quantities were extracted. The *probability of detection* varies strongly with range and non-monotonically: 0.67 below 0.8 m, 0.94 between 1.6 and 2.0 m, and 0.15 beyond 2.0 m. The *monocular range estimator* reads 0.469 of true range with a median absolute error of 1.01 m; the bias is injected at its physical origin, the apparent height, rather than as an additive error on range. The *bearing* error has a systematic component that is a frame convention rather than a sensor property, so only its scatter (0.32°) is injected.

The dataset has a gap between 0.8 and 1.6 m that no amount of care removes, and the detection-probability curve is therefore interpolated across the band in which, as it turned out, the physical runs spent most of their time.

4.3 S2: rate and the structure of the gaps

S2 was fitted to the timing of the same recorded stream. Observations arrive at 3.5 Hz; the marginal acceptance is 0.259 of cycles; and the misses are strongly

clustered, with a mean blind run of 9.16 cycles against the 3.53 that independent dropout would give, and a longest blind run of 57 cycles, about 16 s. That clustering is the point of the level. A planner that loses sight of an obstacle for one cycle at a time behaves quite differently from one that loses it for sixteen seconds, at identical average detection rates.

4.4 S3: the vehicle, downstream of the planner

S3 is a plant model applied to `/planner/cmd_vel_safe`, never to the planner. It was identified from open-water trials: full-command speeds of 0.145 m s^{-1} in surge and 0.123 m s^{-1} in sway; yaw response of $0.0654^\circ \text{ s}^{-1}$ per count with a 383-count deadband in one direction and 0.0537 with 220 in the other; and first-order rise and decay constants of 2.0 s/1.0 s in surge and 0.30 s/0.97 s in sway. The translation axes are reported as symmetric because the trials cannot distinguish their directions at this sample size, not because symmetry was verified; `sway+` rests on one valid trial; the rise constants have spreads as large as their values; and command latency was never measured, so the plant leaves it at zero rather than injecting a plausible value.

4.5 What the ladder does not contain

Two things measured during the campaign are deliberately absent. Pool recirculation was measured at 0.037 m s^{-1} to 0.053 m s^{-1} and was non-stationary enough that successive measurements pointed in opposite directions; fitting a disturbance model to four inconsistent measurements would be invention, so the disturbance is an uncontrolled input present in the physical runs and absent from every simulated one. And, most consequentially, *the ladder contains no visual detector*. Simulated observations come from an oracle projection of the obstacle geometry, degraded by the S1 and S2 statistics; every frozen run records its perception source as an oracle relay and not a visual result. The statistics were fitted to the real detector, so the surrogate is a principled one, but it is a surrogate, and Section 6 reports what the real detector did that no statistical summary of it predicted.

5 The Frozen Simulation Study PRE

5.1 Design and freeze

Eighty closed-loop runs were executed before any physical run: four calibration levels, two planners, two obstacle geometries (K0, obstacle on the approach line; K1, offset 0.35 m), five repetitions per cell. The scenario constants shared by both domains – nominal surge 0.15 m s^{-1} , engagement range 1.5 m, assumed obstacle height 0.5 m – are read by both the simulated campaign and the physical pipeline from a single file, so the two cannot drift apart silently. The aggregate was hashed and committed; we verified during the final audit that all 80 runs still reproduce the committed aggregate exactly, and that every run carries the sentinel of its own calibration level with no leakage between levels.

Table 1. Frozen simulated campaign PRE, median of five repetitions, for the three quantities the text uses. *clearance*: minimum obstacle clearance (m); *span*: manoeuvre span (s); *qualified*: range at the first qualified observation (m). Lateral excursion and commitment distance are in the accompanying repository; the dynamic-window commitment distance equals the starting distance at every level, so it is a left-censored bound rather than a measurement.

geo	planner	clearance				span				qualified			
		S0	S1	S2	S3	S0	S1	S2	S3	S0	S1	S2	S3
K0	committed	1.48	2.22	0.77	0.58	65	56	56	51	3.43	3.03	1.15	1.17
K0	dwa	2.14	2.47	0.93	0.86	60	86	89	92	3.47	3.43	1.23	1.21
K1	committed	1.19	2.56	0.87	0.74	57	55	56	49	3.45	3.21	1.18	1.17
K1	dwa	2.46	2.33	0.87	0.65	73	31	90	92	3.50	3.09	1.21	1.21

5.2 What the levels do to closed-loop behaviour

Table 1 reports the median over five repetitions for each cell. Three patterns are worth stating.

The ladder is not monotone in safety. Minimum clearance for the committed planner in K0 rises from 1.48 m at S0 to 2.22 m at S1, then falls to 0.77 m at S2 and 0.58 m at S3. Adding a measured observation model made the simulated vehicle *safer*, because a biased range estimate that reads short triggers engagement earlier; adding measured timing then removed most of that margin, because the burst structure delays qualification until the obstacle is close. A ladder that simply degraded performance step by step would have been easier to report and would have told us less.

Qualification, not detection, is what the timing model costs. The range at the first planner-valid observation collapses from about 3.4 m at S0 and S1 to about 1.2 m at S2 and S3 in every cell. The vehicle is not blind at long range; it is unable to accumulate enough coherent evidence to be allowed to act.

The two planners respond to calibration in opposite directions. The committed planner’s manoeuvre shortens from 65 s to 51 s between S0 and S3; the dynamic-window planner’s lengthens from 60 s to 92 s. This was the sharpest testable claim in the frozen set, and it does not depend on absolute agreement with anything.

5.3 Failures inside the frozen set

One run of eighty ends in a collision (S2, K0, committed; clearance 0.22 m). Four dynamic-window runs – two at S0, two at S3 – have no commitment distance at all, so those cells are medians of four; the same left-censoring that afflicts the physical runs (Section 7) is already present here, for the same reason, in a planner that begins manoeuvring immediately. Two S2 runs never delivered a single planner-valid observation.

Table 2. The deployment gap REAL: every difference between the stack that produced the frozen predictions and the stack that flew. *forced* marks a change made deliberately on the day; *unnoticed* marks a difference nobody acted on, which the post-deployment ablation (Section 8) finds to matter more than most of the forced ones.

quantity	frozen	flown	status
qualifier warm-up updates	20	2	forced
qualifier confirm updates	3	1	forced
risk enter / exit threshold	0.55 / 0.30	0.30 / 0.15	forced
minimum avoidance hold	1.0 s	4.0 s	forced
engagement range	1.5 m	1.8 m	unintended
observation source	oracle relay + S1/S2	image detector, blob fallback	structural
approach distance	3.5 m	1.86 m	structural
observable range window	unbounded	0.29 m to 2.11 m	unnoticed
clearing offset	2.5 m	2.5 m (basin 2.7 m)	unnoticed
run-past margin	4.0 m	4.0 m (basin 6 m)	unnoticed
run duration	90 s to 120 s	25 s	forced

5.4 Reporting the estimator work

An earlier phase of this project compared four temporal estimator variants under the same qualification. We keep that result only in the compressed form it deserves here: the fixed-gain filter used throughout this paper was selected on that basis, and the choice is held constant in every run reported, so no result below depends on it. Readers interested in the comparison will find it in the accompanying repository; giving it more room would compete with the paper’s one story.

6 Deployment, and the Gap It Exposed

6.1 What happened

The frozen stack was taken to the pool and did not act. In the first runs the detector reported the anchor in a majority of frames, the qualifier discarded every message, and the planner issued no lateral command while the vehicle drove past the obstacle. Over one morning the stack was changed until it produced repeatable lateral responses. The pool was available for that day only. Table 2 is the complete list of what differed by the time the campaign ran, each entry verified against both source files rather than recalled.

6.2 The forced changes, and why each was necessary

Qualification. The warm-up requires twenty mutually coherent updates. Simulated perception delivers a coherent detection every tick and reaches twenty in a second; the real detector produced a box that moved between frames, so twenty never accumulated and the qualifier discarded all 1152 messages of a run while

the vehicle drove past the anchor. Two updates and one confirmation are what the real evidence supports.

Risk threshold. The risk score combines confidence, centrality and apparent area. The simulated obstacle is a broad projection returned at full confidence; the real anchor is a thin shank whose apparent area is small, so the score stayed far below 0.55 even with the anchor detected in 79 % of frames and 75 % of those qualified. Lowering the threshold to 0.30 made the planner act; 0.05, tried first, made it react to the pool walls.

Avoidance hold. A one-second hold was tuned where the simulated vehicle strafes at 0.20 m s^{-1} to 0.30 m s^{-1} . The real vehicle reaches 0.123 m s^{-1} , so one second buys 12 cm and the anchor is still there. Four seconds buys about half a metre. This is the same shortfall that the S3 plant model injects in simulation, appearing in the water.

6.3 The differences nobody noticed, which mattered more

The detector has a range window, and the experiment fitted inside it. The fallback detector accepts a candidate only if its image height lies between 0.15 and 0.90 of the frame. Inverting Equation 1 at the 90° vertical field of view the planner actually used, those gates are an observable range window of 0.29 m to 2.11 m. The vehicle started 1.86 m from the anchor, and the engagement range was 1.80 m: six centimetres apart. The physical planner was inside engagement range at the instant of release. Nothing in the physical campaign measures *when* the planner decides to act, because the answer was fixed by the geometry of the basin and the acceptance gates of the detector before the vehicle moved.

That the approach was 1.86 m was not a choice either. The overhead camera used to measure the start pose covers $4.24 \text{ m} \times 2.38 \text{ m}$, so 1.86 m is the furthest the vehicle can start and still be wholly visible – and a start that cannot be measured before release makes a pre-registered start region meaningless.

The manoeuvre does not fit the basin. The committed planner strafes to a 2.5 m clearing offset and runs 4 m past the obstacle before returning. The basin is 2.7 m to 3.0 m wide, and the anchor is 1.86 m ahead of the start. Neither figure fits. Both were left at their frozen values throughout the campaign, because the deployment effort was directed at the parameters the calibration work had made salient. One run ended against the far wall after a correctly executed avoidance.

The engagement range was never what the record said. The session record states 5.0 m, with a justification for engaging on sight. The runs used 1.8 m: the wrapper that launched every run forwards three arguments and the engagement range was the fourth, so it silently fell back to the runner’s default. All nine runs record 1.8 m. We report this because it is exactly the class of error a matched sim-to-real study is supposed to catch and, in this instance, only the final audit did.

6.4 The planner that could not be flown

The dynamic-window planner issued no command in the physical pipeline. It evaluates route adherence and projects candidate velocities forward, both of which need a pose estimate; in simulation this comes from dead-reckoned odometry that the physical vehicle, with no velocity log and no acoustic positioning, cannot produce.

The evidence deserves stating, because four of the six attempted dynamic-window runs do contain a command stream and could be mistaken for the planner working. Those streams take exactly three values in surge, $\{0, 0.08, 0.15\}$, and three in sway, $\{-0.2, 0, +0.2\}$ – which are the committed planner’s nominal surge, its reduced surge during avoidance, and its strafe setpoint. A dynamic-window planner scores sampled velocity candidates and emits a continuum; a three-level output is a finite-state controller’s. Those commands came from orphaned committed-planner processes surviving on the same topic, the same contamination that the admission criterion of Section 7 exists to catch. The two runs with no orphan present contain a command stream that is identically zero, and that is the dynamic-window planner’s actual output on this vehicle.

The consequence is not subtle and we do not soften it: half of the frozen design was never tested, the physical campaign is single-planner, and every comparison between the two planners in this paper is simulated.

7 Physical Results REAL

7.1 What was flown, and what was recorded

Nine runs were flown on a single day, all with the committed planner. Run 5 is excluded by the criterion described below, leaving eight admissible runs across three geometries: K0 with the anchor on the approach line ($n = 4$), K1 offset 0.35 m ($n = 2$), and K1M, the mirror of K1 ($n = 2$). K1M was invented during the session and was never pre-registered; its purpose and its limits are stated in Section 7.4.

Each run recorded the command stream at 19 Hz and an annotated onboard video. *No external tracking was recorded during any run.* The overhead camera was used to measure and verify the start pose before each release and was deliberately not opened inside the run process, because doing so had killed several runs outright. The consequence is permanent and applies to every physical run in this paper: trajectory, path length, lateral excursion, forward progress, true range to the anchor and true minimum clearance do not exist and cannot be reconstructed.

We could have integrated the commanded velocity and produced all six. We do not, here or anywhere else, because integrating an open-loop model of a saturating, deadbanded vehicle in moving water would manufacture precisely the quantities under test. Where a quantity was not observed, the tables below say so.

Table 3. The eight admissible physical runs REAL. *det*: fraction of frames in which the detector reported the obstacle; *surge*: mean commanded surge; *lat*: fraction of samples with a lateral command; *rev*: sign reversals; *side* and t_c : side and time of the first sustained lateral commitment ($> 0.05 \text{ m s}^{-1}$ held 1.0 s). Entries marked *cens.* began already committed, so the time is a bound. Commitment *distance* is absent from this table because it was never observed.

run	geo	Hz	det	surge	lat	rev	side	t_c (s)
1	K0	18.9	0.79	0.149	0.32	1	right	8.84
2	K0	18.8	0.89	0.113	0.53	2	right	5.16
3	K0	36.2	0.88	0.133	0.24	0	right	cens.
4	K0	55.6	0.97	0.113	0.53	2	right	cens.
6	K1	18.7	0.94	0.105	0.64	1	right	6.05
7	K1	18.7	0.85	0.123	0.38	1	left	10.53
8	K1M	18.6	0.84	0.120	0.43	1	left	6.58
9	K1M	18.7	0.71	0.110	0.57	3	left	8.53

7.2 Admissibility

Some recordings contain more command samples than the 19 Hz publication grid holds, because orphaned planner processes from earlier runs survived a name-based process sweep and published to the same topic. Where the surplus samples are exact duplicates they carry no information and collapsing them onto the grid is resampling; where they are not, they are genuine divergence between publishers and the run cannot be repaired. The admission test is measured rather than chosen: the threshold is the largest within-cell command spread among recordings made *at* the grid rate, which have a single publisher by construction (0.0107 m s^{-1}). Eight runs pass; run 5, at 0.0349 m s^{-1} , does not, and is excluded while its recording is retained.

7.3 Command-domain results

Table 3 reports every admissible run individually. With eight runs and two per off-axis geometry, we report runs and medians and avoid inferential statistics entirely; an effect size computed on $n = 2$ would be decoration.

Three observations. The vehicle detected the anchor in 71 % to 97 % of frames, and no collision was recorded in an admissible run; commanded surge is zero in less than 1 % of samples throughout. Median commanded surge is 0.123, 0.114 and 0.115 m s^{-1} in K0, K1 and K1M – propulsion does not depend on geometry, as it should not. And three of the four K0 runs are left-censored: they were already commanding laterally at the first recorded sample, which follows directly from starting 6 cm outside the engagement range.

7.4 Side choice follows the obstacle

Through the first seven runs the vehicle chose the same side six times. Two explanations fit that equally well – the planner selects its side from where it sees

the obstacle, or the vehicle simply always goes right for a mechanical, perceptual or tie-breaking reason – and neither K0 nor K1 can separate them, because K0 puts the anchor dead ahead and K1’s offset lies on the side a rightward bias would already favour. Mirroring the offset separates them in a single run, so the geometry was mirrored and two runs flown.

Both went left, against four of four going right in K0. Read with its sample size, this is consistent with side choice following the obstacle and inconsistent with a fixed bias, and it is the strongest statement two runs can carry. K1M was not pre-registered, was designed after seeing six same-side outcomes, and has $n = 2$; we treat it as an exploratory observation that motivated the simulated study in Section 8, not as a confirmatory result.

7.5 The recovered perception stream

The detector drew its own output on every frame it processed – a green box for an accepted detection, orange for a rejected one, nothing when it saw nothing – so the observation stream that reached the planner can be read back exactly from the recordings, for the six runs that have video. This is recovery, not re-detection: no detector is re-run, and the numbers are what the planner was actually given. Reading the box height through Equation 1 recovers the range the planner *believed*, which is the quantity that exists on both sides of the sim-real comparison. The recovery’s own measurement error, from the four-pixel stroke width, has a median of 0.02 m, two orders below the estimator’s own bias.

The recovered stream shows what the range window of Section 6 implies. Accepted detections occupy 71 % to 91 % of frames and their median believed range is 0.94 m to 1.51 m; between 0 and 28 % of them sit exactly at the detector’s upper size gate, where the reported range saturates at 0.30 m. We report that saturation fraction rather than a minimum range, because the minimum is the gate and not the anchor.

8 Post-Deployment Diagnostic Simulation POST

8.1 Design and provenance

All simulations in this section were designed and run after the physical campaign, with its outcome known. They are diagnostic evidence (POST), never pre-deployment predictions.

The frozen 80-run aggregate is unsuitable as an ablation baseline because its stochastic seed is not reproducible and its loaded execution rate differs from an idle rerun (Section 10). We therefore ran a self-contained 80-run ablation: eight configurations, ten repetitions each, all using the committed planner, K0 and S3. A0 reproduces the frozen configuration in the same session. A1–A6 each change only the listed factor relative to A0; A7 combines A1–A6. Comparisons use medians, interquartile ranges and counts because the outcomes are bimodal.

A4 uses the 1.8 m value in every run record, not the erroneous 5.0 m session file. A5 measures the relay defect found only in the final audit. A6 tests one

Table 4. Diagnostic configurations POST. A1–A6 are single-factor changes from A0. A7 combines them and was not flown.

rung change		provenance
A0	frozen configuration	same-session baseline
A1	qualifier 20/3 $\rightarrow$ 2/1 updates	forced in pool
A2	risk 0.55/0.30 $\rightarrow$ 0.30/0.15	forced in pool
A3	hold 1.0 $\rightarrow$ 4.0 s	forced in pool
A4	engagement 1.5 $\rightarrow$ 1.8 m	executed value
A5	relay FOV 60 $\rightarrow$ 90°	post-session audit fix
A6	offset 2.5 $\rightarrow$ 0.6 m, run-past 4.0 $\rightarrow$ 1.0 m	diagnostic; not flown
A7	A1–A6 together	combined diagnostic; not flown

Table 5. Deployment-gap ablation POST, ten runs per configuration. Continuous values are medians with interquartile ranges. *go-around* counts runs reaching the second manoeuvre phase; Table 4 defines each change.

rung change		clear. (m)	lateral (m)	manoeuvred	go-around	coll.
A0	frozen	1.29 [0.42–2.66]	1.67 [1.35–3.96]	8/10	10/10	2
A1	qualifier	2.22 [1.33–3.12]	2.76 [2.21–4.47]	9/10	10/10	2
A2	risk thresholds	1.90 [0.73–2.23]	1.98 [1.29–3.31]	9/10	10/10	2
A3	hold	2.43 [1.79–3.15]	1.73 [1.28–4.23]	9/10	10/10	0
A4	engagement	1.48 [0.30–2.62]	1.52 [1.20–2.06]	10/10	10/10	3
A5	relay FOV fix	2.91 [2.41–3.23]	1.56 [0.03–2.73]	7/10	10/10	0
A6	tested pool rescaling	0.30 [0.28–2.95]	0.47 [0.00–0.82]	7/10	10/10	4
A7	combined diagnostic	3.13 [2.84–3.14]	2.33 [0.85–6.02]	10/10	6/10	2

specific rescaling intended to fit the basin; it is neither an executed configuration nor an exhaustive search over basin-compatible manoeuvres.

8.2 Results

A5 produces the largest single-factor increase in median clearance: 1.29 m to 2.91 m, with zero rather than two collisions. The relay had applied the measured detection-probability curve in the wrong range band. This quantifies a defect in the frozen simulator; it does not revise the frozen results.

A6 yields median clearance 0.30 m and four collisions. The tested pool-feasible rescaling was therefore unsafe. It indicates strong sensitivity to detour width but does not establish that every basin-compatible manoeuvre is unsafe.

A1–A3 individually improve median clearance relative to A0, and A3 has no collisions. A4 is the only executed single-factor change that reduces safety in this simulation, with three collisions. These post-hoc results do not identify causal effects in the physical runs.

A0–A6 reach the go-around phase in all ten runs; A7 does so in six. No physical run reached it. Because A7 includes the post-audit FOV fix and the unflown A6 rescaling, it is a combined diagnostic configuration, not the flown stack. More generally, the open simulated scene contains no basin boundary, so

none of these configurations reproduces the structural constraint that dominated deployment.

9 Simulation-to-Reality Analysis

9.1 Where the two domains can be compared at all

The command stream is recorded on the same topic, `/planner/cmd_vel_safe`, at the same boundary in both domains, and is therefore the one axis on which a simulated and a physical run measure the same signal with the same instrument. Everything spatial is excluded by construction (Section 7).

What follows is not a test of predictive validity. The frozen and the flown stack differ in the eleven ways of Table 2, so agreement would not establish that the calibration worked and disagreement would not establish that it failed. The question is narrower: how far apart are the two domains where both measure the same thing, and does the distance move with the calibration level?

9.2 The calibration ladder does not move the command domain

Table 6 answers the second half first. Median commanded surge in simulation is 0.1428 to 0.1438 m s^{-1} at every level and in both geometries – a spread of one part in a hundred and forty across the entire ladder. The physical median is 0.123 m s^{-1} in K0. The absolute difference is therefore 0.0198 m s^{-1} to 0.0207 m s^{-1} at S0, S1, S2 and S3 alike, and it does not shrink as the simulator becomes more realistic.

This is the expected result once stated, and it is worth stating because it bounds what a boundary-level comparison can ever show. S3 acts downstream of this signal by design – that is what makes it a plant model and not a controller edit – so it cannot move the commanded surge at all. S1 and S2 act upstream, but they change *when* the planner acts rather than how hard it pushes, and the planner’s surge setpoints are constants. A study that reported shrinking command-domain error across such a ladder would be reporting something other than what it claimed.

9.3 The manoeuvre that never happened

The informative disagreement is not in a median but in a threshold. The committed planner strafes at 0.20 m s^{-1} while it displaces to its clearing offset, and commands 0.30 m s^{-1} of sway with 0.40 m s^{-1} of surge only once it enters the go-around phase and begins running past the obstacle. The peak lateral command therefore says, unambiguously and without any spatial measurement, whether a run ever reached that phase.

Every one of the twenty frozen simulated runs in K0 reached it, at all four calibration levels. *None of the eight physical runs did.* All eight peak at exactly 0.200 m s^{-1} of lateral command and exactly 0.150 m s^{-1} of surge – the strafe setpoint and the nominal surge, never the go-around values.

Table 6. Command-domain comparison, committed planner. Simulated values are medians over five frozen runs per cell PRE; physical values are medians over the admissible runs of each geometry REAL($n = 4, 2, 2$). *peak* is the largest lateral command: the planner strafes at 0.20 m s^{-1} and reaches 0.30 m s^{-1} only after entering its go-around phase.

domain	geo	n	surge (m s^{-1})	lateral frac.	peak (m s^{-1})	reversals	
sim	S0	K0	5	0.143	0.34	0.300	2
sim	S1	K0	5	0.144	0.30	0.300	2
sim	S2	K0	5	0.143	0.34	0.300	2
sim	S3	K0	5	0.143	0.30	0.300	2
real	K0	4	0.123	0.42	0.200	1.5	
real	K1	2	0.114	0.51	0.200	1	
real	K1M	2	0.115	0.50	0.200	2	

The physical vehicle, in other words, spent every run in the first phase of a manoeuvre it never completed. That is precisely what the geometry of Section 6 predicts: the clearing offset is 2.5 m in a basin 2.7 m to 3.0 m wide, so the displacement condition that ends the strafe phase can never be satisfied, and the planner strafes until the run ends or the operator intervenes. The higher physical lateral fraction (0.42 to 0.51 against 0.30 to 0.34 in simulation) is the same fact seen from another side: more of the run is spent strafing because the strafe never terminates.

We regard this as the most useful sim-real finding in the study, and note what made it available. It required no ground truth, no tracking and no spatial reconstruction – only the recognition that a controller’s own setpoint constants make its internal phase observable in the command stream. Where external reference is unavailable, as it is on most small vehicles most of the time, instrumenting the controller’s phase boundaries in this way recovers a genuine behavioural comparison from data that otherwise supports only summary statistics.

9.4 Commitment: measurable in one domain, predetermined in the other

Commitment distance is absent from the physical results for a reason beyond missing instrumentation. The vehicle was inside engagement range at release, and three of the four K0 runs are already commanding laterally at their first sample, so even with perfect tracking the physical commitment distance would be a property of where the vehicle was put rather than of what the planner decided. A sim-real error on that quantity would compare a measurement against a geometric certainty. Commitment *time* remains meaningful and is given in Table 3, censored where the run began already committed.

10 Limitations

We separate limitations of the physical campaign, defects we found in our own frozen study, and limits of the calibration itself. The second group is unusual to publish; we publish it because a frozen result whose flaws are known privately is worth less than one whose flaws are stated.

10.1 The physical campaign

Eight runs, one planner, one pool and one day. The study used one static anchor. The design called for twenty runs across two planners; the dynamic-window planner was never flown. K1 and the mid-session K1M have two runs each, and no confirmatory claim rests on a physical difference between geometries. The deployed configuration was also modified during the session and differs from the frozen stack.

No external ground truth. Nothing spatial about the physical runs was measured. This most constrains the paper and also produces one of its results: a camera-only field campaign of this kind can validate a stack’s command and perception domains and not its trajectory. Record external tracking during the runs, not only before them.

Uncontrolled disturbance. Pool recirculation was measured at 0.037 m s^{-1} to 0.053 m s^{-1} , of the same order as a third of the vehicle’s commanded surge, and was non-stationary. It is present in every physical run and absent from every simulated one.

Orphaned processes contaminated part of the session. Duplicate planner instances survived process cleanup and published to the command topic, which is why the admission criterion of Section 7 exists. Three admissible runs were recorded at above-grid rates and are admitted on the measured-duplicate test; run 5 is excluded. We regard the criterion as defensible but note that it was formulated after the data existed, which is exactly the circumstance in which a threshold can be tuned to a desired answer. Ours is derived from the recordings that had a single publisher by construction, and we state it so a reader can disagree.

10.2 Defects in the frozen study

The observation relay and the planner disagree about the camera. The relay decides whether a detection survives by converting apparent height to range at a 60° vertical field of view, while the oracle projects and the planner inverts at 90° . The relay therefore believes the obstacle roughly 1.5 times further away than the planner does, and applies the measured detection-probability curve to the wrong range band, in the sixty runs at S1 and above. The frozen results are what they are and we do not restate them; the effect is quantified as one rung of the post-deployment ablation.

The frozen runs are not reproducible, for two independent reasons. The relay’s random seed is discarded when it is zero, the value the campaign used, so the

stochastic levels were seeded from system entropy. And the campaign’s command loop ran at about half its intended rate – 10 Hz on the nominal topic against 20 Hz on an idle machine – because that loop is wall-clock driven and twenty runs ran back to back on a loaded machine, while the perception gate is simulated-time driven and was unaffected.

A benchmark constant that nothing reads. The frozen benchmark records a 60° vertical field of view for the shared range estimator. No node in either domain reads it; both used the planner’s 90° default. The domains therefore agree and no sim-real mismatch follows, but a file whose purpose is to keep them aligned did not enforce this value.

Individual run failures. One frozen run of eighty ends in a collision; two S2 runs never produced a planner-valid observation; four dynamic-window runs have no commitment distance, making those cells medians of four.

10.3 The calibration

The paired-observation dataset has no coverage between 0.8 and 1.6 m, which is the band the physical runs spent most of their time in. Command latency was never measured and is left at zero rather than guessed. Rise-time constants have spreads as large as their values. The translation axes are reported as symmetric because the sample cannot distinguish their directions, not because symmetry was established, and `sway+` rests on a single valid trial. Two actuator trials ended in contact and are excluded from the fit with their reason recorded.

Most fundamentally, the ladder models the *statistics* of the real detector and not the detector. The frozen study never ran a visual detector at all, and the property of the real detector that most shaped the physical campaign – the range window imposed by its acceptance gates – is not a statistic that any of S1, S2 or S3 could have contained.

Post-hoc scope. All 80 ablation runs were designed after the physical outcome was known. They measure sensitivity in simulation, not physical causal effects. A6 tests one pool-feasible rescaling rather than all compatible manoeuvres, and A7 combines the post-audit FOV correction with that unflown rescaling; A7 is not the flown configuration. The frozen scene also contains no basin boundary.

10.4 What we would do differently

Record external tracking during runs, not only before them. Derive the scenario geometry from the basin rather than adapting the basin to a scenario. Check that every parameter in a shared benchmark file is read by something. Seed every stochastic component explicitly and assert the seed at run time. And treat the acceptance gates of a perception module as a first-class part of its interface, because in this study they determined the shape of the experiment more than any quantity we calibrated.

11 Conclusion

We set out to test whether measuring a simulator’s mismatches against the real system makes its closed-loop predictions transfer, and we committed the predictions before going to the water so that the test would be a test. The stack did not transfer, and the two differences that shaped the physical experiment most were not calibrated quantities at all: the detector’s acceptance gates, and the size of the room.

The honest reading is that our calibration effort was aimed at the quantities that were easy to measure rather than the ones that governed the outcome. Detection probability, range bias, dropout burstiness and actuator response are all real, all were measured against the physical vehicle, and all were faithfully injected; none of them determined whether the system worked in the pool.

The post-deployment ablation sharpened that reading in two ways we did not expect. The largest single effect on simulated safety was not a calibration level or a field change but a defect in our own frozen simulator: correcting the relay’s field of view raised median clearance from 1.29 m to 2.91 m and removed its collisions. The specific pool-feasible rescaling tested in A6 reduces median clearance to 0.30 m and produces four collisions in ten runs. This shows strong sensitivity to detour width; it does not establish that every basin-compatible manoeuvre would be unsafe.

We think four things generalise. First, the acceptance gates of a perception module belong in its published interface, because they define the range interval over which it produces any observation at all, and that interval can silently exclude the experiment. Second, a sim-to-real study should derive its scenario geometry from the physical site before freezing anything, since a manoeuvre the site cannot contain turns a controller result into an architecture result. Third, the boundaries of the physical workspace belong in the simulated scenario: ours had none, which is why all twenty corresponding frozen runs reached a manoeuvre phase that no physical run reached, and no calibration of sensing or actuation could have supplied that. Fourth, and most uncomfortably, a campaign that adapts a stack in the field has stopped testing the stack it froze; saying so plainly costs a claim and buys back the reader’s ability to judge everything else.

The physical campaign supports a narrower claim. The controller operated in closed loop during eight admissible BlueROV2 runs; no collision was recorded. The command log shows geometry-independent surge and image-relative side selection, the latter based on two mirrored-geometry runs and therefore exploratory. It cannot support, and we do not claim, causal validation of the calibration ladder, any ranking of the two planners in water, and anything at all about the vehicle’s trajectory, which was never measured.

The pool is drained and the physical dataset is closed. The diagnostic simulations were run afterwards, with the outcome known, and are labelled accordingly. Their role is to expose simulated sensitivity, not to turn a post-hoc diagnosis into a prediction.

References

1. Bergantin, L., Viel, C., Jaulin, L.: Real-time braking control based on optic flow divergence onboard an underwater vehicle. *Ocean Engineering*, vol. 310, art. 118674 (Elsevier), October 2024 (2024). <https://doi.org/10.1016/j.oceaneng.2024.118674>
2. Drews-Jr, P., Hernández, E., Elfes, A., Nascimento, E.R., Campos, M.F.M.: Real-time monocular obstacle avoidance using underwater dark channel prior. *IEEE/RSJ IROS 2016* (2016). <https://doi.org/10.1109/IROS.2016.7759687>
3. Eriksen, B.O.H., Breivik, M., Pettersen, K.Y., Wiig, M.S.: A modified dynamic window algorithm for horizontal collision avoidance for AUVs. In: *Proc. IEEE Conf. Control Applications (CCA)*. pp. 499–506 (2016). <https://doi.org/10.1109/CCA.2016.7587879>
4. Fox, D., Burgard, W., Thrun, S.: The dynamic window approach to collision avoidance. *IEEE Robotics & Automation Magazine* 4(1), 23–33 (1997). <https://doi.org/10.1109/100.580977>
5. Kadian, A., Truong, J., Gokaslan, A., Clegg, A., Wijmans, E., Lee, S., Savva, M., Chernova, S., Batra, D.: Sim2real predictivity: Does evaluation in simulation predict real-world performance? *IEEE Robotics and Automation Letters* 5(4), 6670–6677 (2020). <https://doi.org/10.1109/LRA.2020.3013848>
6. Meyers, B., Mangelson, J.G.: Testing and evaluation of underwater vehicle using hardware-in-the-loop simulation with HoloOcean. In: *Proc. OCEANS 2025 – Great Lakes* (2025). <https://doi.org/10.23919/OCEANS59106.2025.11245188>
7. Potokar, E., Ashford, S., Kaess, M., Mangelson, J.G.: HoloOcean: An underwater robotics simulator. In: *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*. pp. 3040–3046 (2022). <https://doi.org/10.1109/ICRA46639.2022.9812353>
8. Potokar, E., Lay, K., Norman, K., Benham, D., Ashford, S., Peirce, R., Neilsen, T.B., Kaess, M., Mangelson, J.G.: HoloOcean: A full-featured marine robotics simulator for perception and autonomy. *IEEE Journal of Oceanic Engineering* 49(4), 1322–1336 (2024). <https://doi.org/10.1109/JOE.2024.3410290>
9. Song, J., Ma, H., Bagoren, O., Sethuraman, A.V., Zhang, Y., Skinner, K.A.: Ocean-sim: A gpu-accelerated underwater robot perception simulation framework. *IROS 2025*; arXiv:2503.01074 (2025), <https://arxiv.org/abs/2503.01074>
10. Truong, J., Rudolph, M., Yokoyama, N.H., Chernova, S., Batra, D., Rai, A.: Rethinking sim2real: Lower fidelity simulation leads to higher sim2real transfer in navigation. In: *Proc. Conf. Robot Learning (CoRL)*, PMLR vol. 205. pp. 859–870 (2022), <https://proceedings.mlr.press/v205/truong23a.html>
11. Yang, P., Liu, H., Roznere, M., Li, A.Q.: Monocular camera and single-beam sonar-based underwater collision-free navigation with domain randomization. *International Symposium on Robotics Research (ISRR) 2022*, Springer Proceedings in Advanced Robotics (2022), <https://arxiv.org/abs/2212.04373>
12. Yao, P., Sui, X., Liu, Y., Zhao, Z.: Vision-based environment perception and autonomous obstacle avoidance for unmanned underwater vehicle. *Applied Ocean Research*, vol. 134, art. 103510 (Elsevier) (2023). <https://doi.org/10.1016/j.apor.2023.103510>
13. Yu, B., Wu, J., Islam, M.J.: Udepth: Fast monocular depth estimation for visually-guided underwater robots. *IEEE ICRA 2023* (2023). <https://doi.org/10.1109/ICRA48891.2023.10161471>
14. Zhang, S., Zhang, L., Chen, Y.: Sim-to-real pipeline for training autonomous obstacle avoidance of underwater robots based on high-fidelity model. *Robotica* 43(8), 2952–2974 (2025). <https://doi.org/10.1017/S0263574725101999>